13/04/26

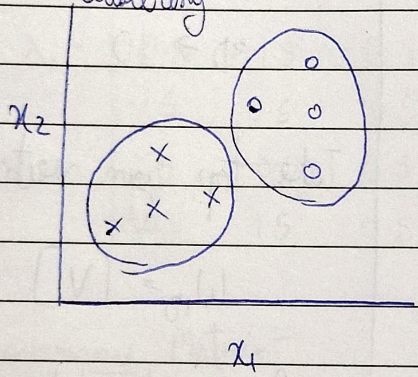
* Unsupervised learning :-

~~Problems~~

1. Visualization → tSNE (non-linear projectⁿ)
2. Grouping / clustering :-

   (i) K-means
   (ii) Hierarchical clustering
   ↳ Agglomerative clustering

* clustering :-

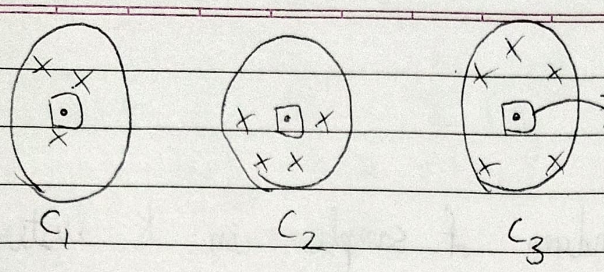


x_1	x_2
2	4.3
3	3.8
4	5.6

* Notations :-

- n samples
- $x^{(i)}$ $i \in 1$ to n , $x \in \mathbb{R}^d$
- $C_1, C_2 \ldots C_K$ denote the sets containing indices of the observations in each cluster.
- K is the total no. of clusters # this is decided by us.

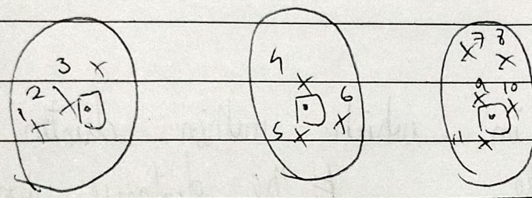
There shld be a notion of similarity in the cluster
eg. students of cluster A will belong to the same state



→ centroid (mean)
↳ we calculate this ~~with the~~ by the data points in the cluster.

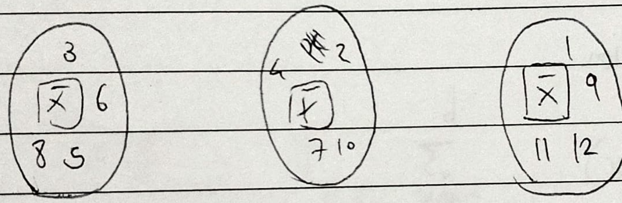
Step 1:

Random assignment



$x^{(1)}, x^{(2)} \dots\dots x^{(n)}$

Step 2:



Compute distance $x^{(1)}$ & centroid$_1$

* Convergence @ when ~~g~~ ~~gradt g~~ the centroid doesn't change

∄ a datapoint will only be in 1 cluster.

$C_1 \cup C_2 \cup C_3 \cup \dots\dots C_K = \{1, 2, 3 \dots$ samples

$C_1 \cap C_2 \neq \phi$

* Clustering computed:

(1) within ~~distance~~ distance ~~pts~~ ~~pts~~ (points)
~~Between~~ ~~pts~~

(2) Btwn cluster distance (points)